

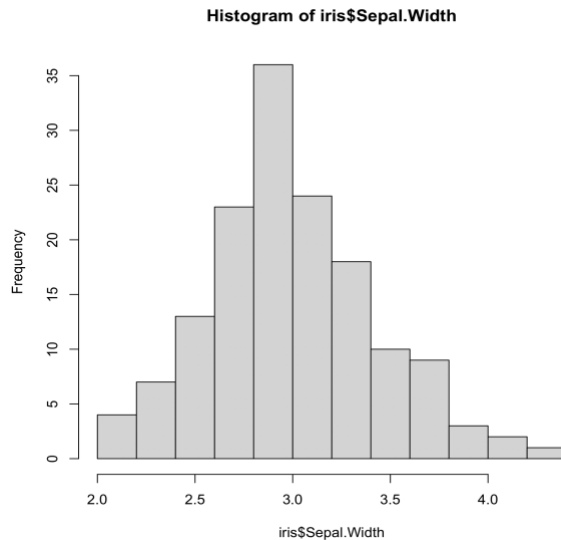
GRAD 50500 – Foundation in Data Science – Week 3 Assignment – Mudassir Hashmi

Iris Dataset:

- a. Make a histogram of the variable Sepal.Width.

Code and Corresponding out:

```
hist(iris$Sepal.Width)
```



- b. Based on the histogram from #1a, which would you expect to be higher, the mean or the median:

- *Mean is the average value, and Median is the middle value. It appears in the Histogram the distribution is slight out hence most likely the mean would probably be higher.*

- c. Confirm your answer to #1b by finding these values:

```
> mean(iris$Sepal.Width)
```

```
[1] 3.057333
```

```
> median(iris$Sepal.Width)
```

```
[1] 3
```

- *Mean is 3.05733 and Median is 3 making Mean slightly higher.*

d. Only 27% of the flowers have a Sepal.Width higher than ____3.3____ cm.

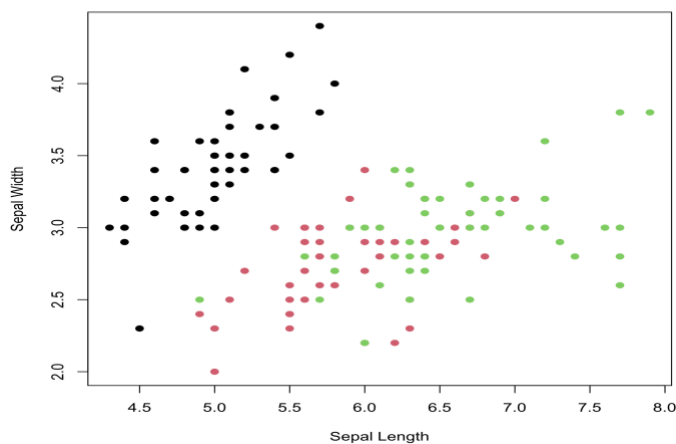
```
> quantile(iris$Sepal.Width, 0.73)
```

73%

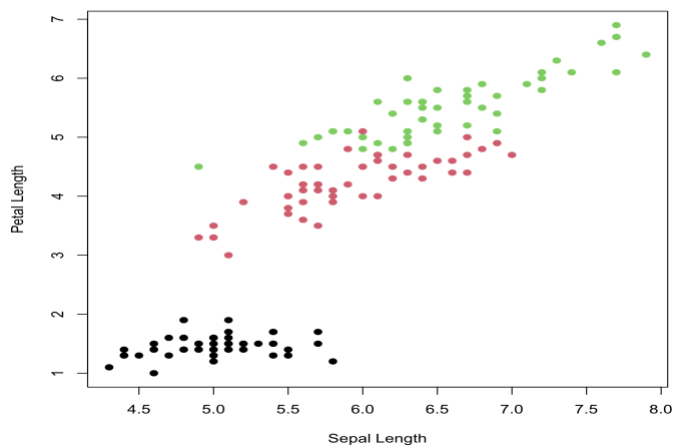
3.3

e. Make scatterplots of each pair of the numerical variables in iris (There should be 6 pairs/plots).

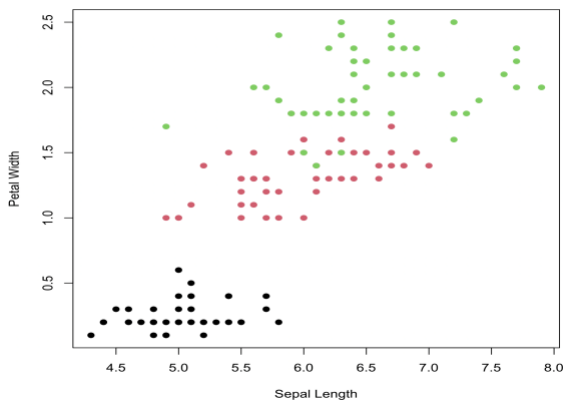
```
> plot(iris$Sepal.Length, iris$Sepal.Width, col=iris$Species, pch=19, xlab="Sepal Length",  
ylab="Sepal Width")
```



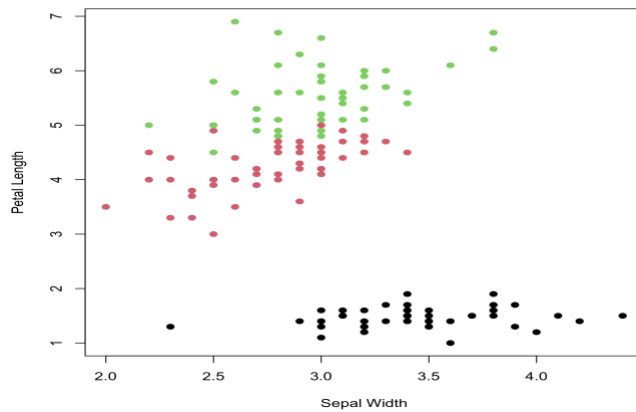
```
> plot(iris$Sepal.Length, iris$Petal.Length, col=iris$Species, pch=19, xlab="Sepal Length",  
ylab="Petal Length")
```



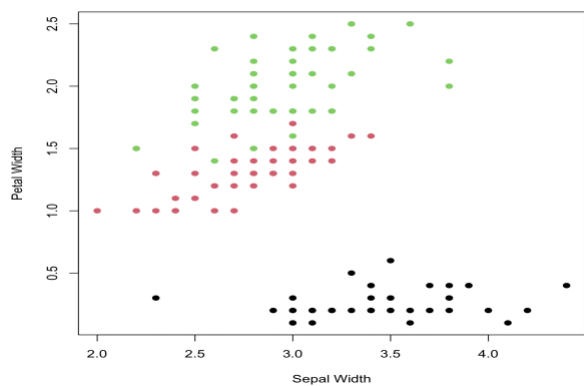
```
> plot(iris$Sepal.Length, iris$Petal.Width, col=iris$Species, pch=19, xlab="Sepal Length",
ylab="Petal Width")
```



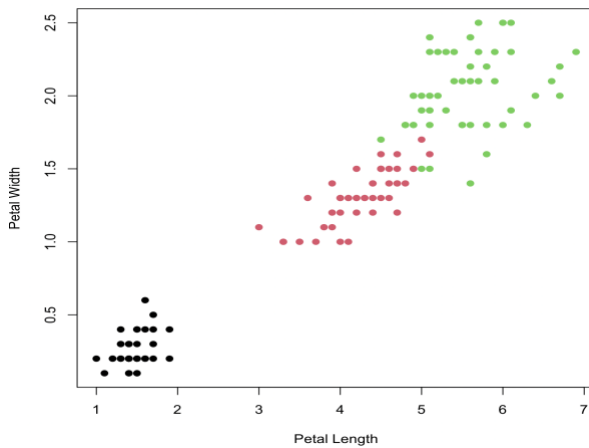
```
> plot(iris$Sepal.Width, iris$Petal.Length, col=iris$Species, pch=19, xlab="Sepal Width",
ylab="Petal Length")
```



```
> plot(iris$Sepal.Width, iris$Petal.Width, col=iris$Species, pch=19, xlab="Sepal Width",
ylab="Petal Width")
```



```
> plot(iris$Petal.Length, iris$Petal.Width, col=iris$Species, pch=19, xlab="Petal Length",
ylab="Petal Width")
```



- f. Based on #1e, which two variables appear to have the strongest relationship? And which two appear to have the weakest relationship?
- *Petal Length and Petal Width is more linear in nature hence most likely has the stronger relationship. Sepal Length vs Sepal Width is more non-linear according to the Scatter plot hence most likely has the weaker relationship.*

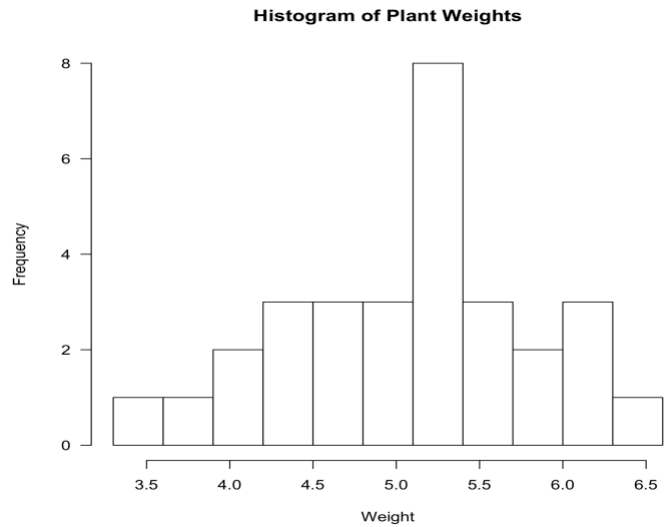
Plant Growth Dataset:

- a. Make a histogram of the variable weight with breakpoints (bin edges) at every 0.3 units, starting at 3.3.

```
> data("PlantGrowth")

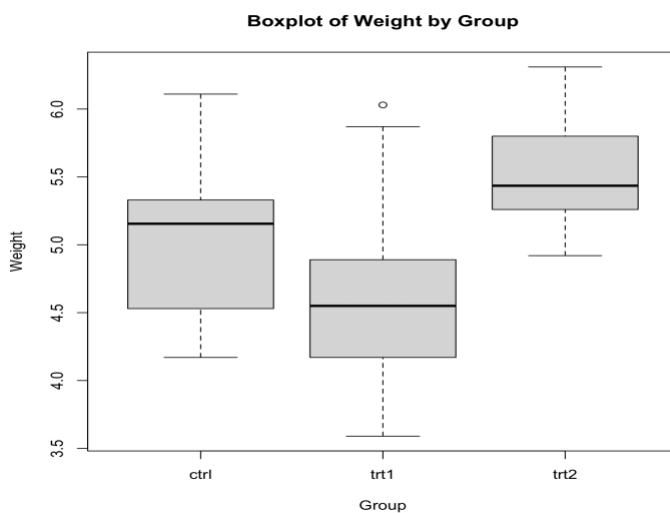
> custom_breaks <- seq(3.3, 6.6, by = 0.3)

> hist(PlantGrowth$weight,
+     breaks = custom_breaks,
+     main = "Histogram of Plant Weights",
+     xlab = "Weight",
+     ylab = "Frequency",
+     col = "white",
+     las = 1)
```



b. Make boxplots of weight separated by group in a single graph.

```
> boxplot(weight ~ group, data = PlantGrowth,
+         main = "Boxplot of Weight by Group",
+         xlab = "Group",
+         ylab = "Weight")
```



c. Based on the boxplots in #2b, approximately what percentage of the "trt1" weights are below the minimum "trt2" weight?

- Based on the Boxplot close to 45%-50% weights are below the minimum trt2 weight.

- d. Find the exact percentage of the "trt1" weights that are below the minimum "trt2" weight.

```
> trt1_weights <- PlantGrowth$weight[PlantGrowth$group == "trt1"]
```

```
> trt2_weights <- PlantGrowth$weight[PlantGrowth$group == "trt2"]
```

```
> mean(trt1_weights < min(trt2_weights)) * 100
```

```
[1] 80
```

- e. Only including plants with a weight above 5.5, make a barplot of the variable group. Make the barplot colorful using some color palette (in R, try running `?heat.colors` and/or check out <https://www.r-bloggers.com/palettes-in-r/>).

```
> filtered_data <- subset(PlantGrowth, weight > 5.5)
```

```
> group_counts <- table(filtered_data$group)
```

```
> barplot(group_counts,
```

```
+   main = "Count of Groups for Plants with Weight > 5.5",
```

```
+   xlab = "Group",
```

```
+   ylab = "Count",
```

```
+   col = rainbow(length(group_counts)), # Colorful palette
```

```
+   border = "black")
```

